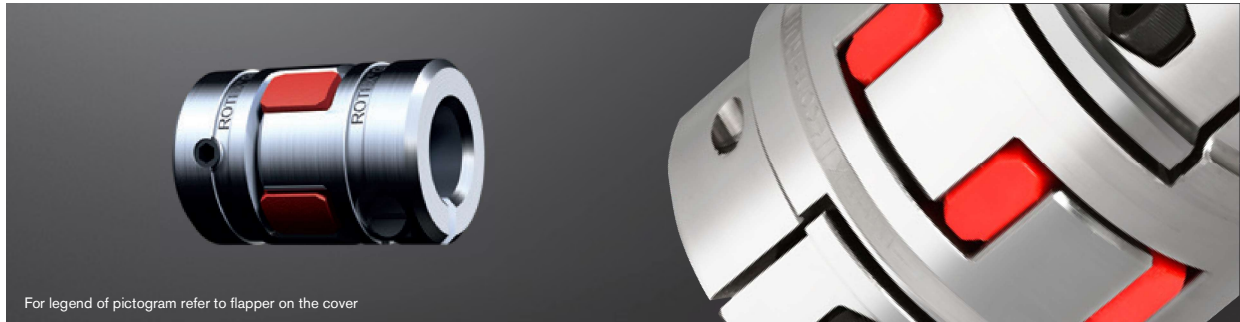


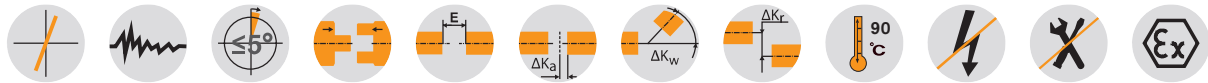
# ROTEX® GS

## Backlash-free jaw couplings

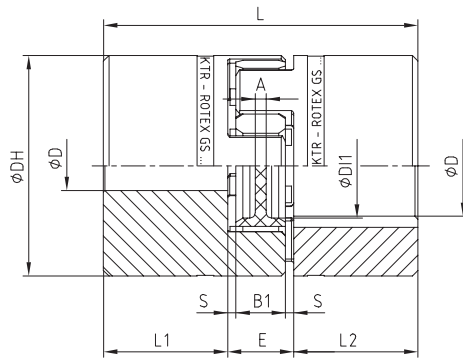
### Standard types



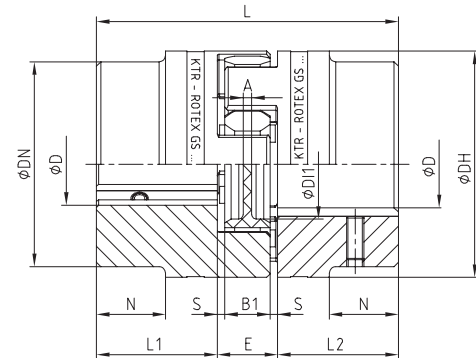
For legend of pictogram refer to flapper on the cover



ROTEX® GS 5 - 38

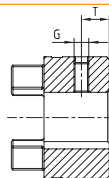


ROTEX® GS 42 - 90



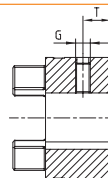
#### Types of hubs:

Type 1.0



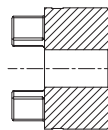
with feather keyway  
and setscrew

Type 1.1



without feather keyway,  
with setscrew

Type 1.2



without feather keyway  
and without setscrew

#### ROTEX® GS standard types - For size 5 to 38 hub material aluminium/for size 42 to 90 hub material steel

| Size | Spider GS <sup>1)</sup><br>torque T <sub>KN</sub> [Nm] |        |        | D <sub>max.</sub><br>for hub type |     |     | Dimensions [mm] |     |      |     |        |      |    |    |     |     |      |     |         | Setscrew DIN EN ISO 4029 |  |  |
|------|--------------------------------------------------------|--------|--------|-----------------------------------|-----|-----|-----------------|-----|------|-----|--------|------|----|----|-----|-----|------|-----|---------|--------------------------|--|--|
|      | 98 ShA                                                 | 57 ShD | 64 ShD | 1.0                               | 1.1 | 1.2 | DN              | DH  | DI1  | L   | L1, L2 | M, N | E  | B1 | S   | A   | G    | T   | TA [Nm] |                          |  |  |
| 5    | 0.9                                                    | –      | –      | –                                 | 6   | 6   | –               | 10  | –    | 15  | 5      | –    | 5  | 4  | 0.5 | 4.0 | M 2  | 2.5 | 0.35    |                          |  |  |
| 7    | 2.0                                                    | –      | 2.4    | 7                                 | 7   | 7   | –               | 14  | –    | 22  | 7      | –    | 8  | 6  | 1.0 | 6.0 | M 3  | 3.5 | 0.6     |                          |  |  |
| 9    | 5.0                                                    | –      | 6      | 10                                | 11  | 11  | –               | 20  | 7.2  | 30  | 10     | –    | 10 | 8  | 1.0 | 1.5 | M 4  | 5.0 | 1.5     |                          |  |  |
| 12   | 9.0                                                    | –      | 12     | 12                                | 12  | 12  | –               | 25  | 8.5  | 34  | 11     | –    | 12 | 10 | 1.0 | 3.5 | M 4  | 5.0 | 1.5     |                          |  |  |
| 14   | 12.5                                                   | –      | 16     | 16                                | 16  | 16  | –               | 30  | 10.5 | 35  | 11     | –    | 13 | 10 | 1.5 | 2.0 | M 4  | 5.0 | 1.5     |                          |  |  |
| 19   | 21                                                     | 23     | 26     | 24                                | –   | –   | –               | 40  | 18   | 66  | 25     | –    | 16 | 12 | 2.0 | 3.0 | M 5  | 10  | 2.0     |                          |  |  |
| 24   | 60                                                     | 68     | 76     | 32                                | –   | –   | –               | 55  | 27   | 78  | 30     | –    | 18 | 14 | 2.0 | 3.0 | M 5  | 10  | 2.0     |                          |  |  |
| 28   | 160                                                    | 180    | 200    | 38                                | –   | –   | –               | 65  | 30   | 90  | 35     | –    | 20 | 15 | 2.5 | 4.0 | M 8  | 15  | 10      |                          |  |  |
| 38   | 325                                                    | 365    | 405    | 45                                | –   | –   | –               | 80  | 38   | 114 | 45     | –    | 24 | 18 | 3.0 | 4.0 | M 8  | 15  | 10      |                          |  |  |
| 42   | 450                                                    | 495    | 560    | 55                                | –   | –   | 85              | 95  | 46   | 126 | 50     | 28   | 26 | 20 | 3.0 | 4.0 | M 8  | 20  | 10      |                          |  |  |
| 48   | 525                                                    | 590    | 655    | 62                                | –   | –   | 95              | 105 | 51   | 140 | 56     | 32   | 28 | 21 | 3.5 | 4.0 | M 8  | 20  | 10      |                          |  |  |
| 55   | 685                                                    | 770    | 825    | 74                                | –   | –   | 110             | 120 | 60   | 160 | 65     | 37   | 30 | 22 | 4.0 | 4.5 | M 10 | 20  | 17      |                          |  |  |
| 65   | 940                                                    | 1055   | 1175   | 80                                | –   | –   | 115             | 135 | 68   | 185 | 75     | 47   | 35 | 26 | 4.5 | 4.5 | M 10 | 20  | 17      |                          |  |  |
| 75   | 1920                                                   | –      | 2400   | 95                                | –   | –   | 135             | 160 | 80   | 210 | 85     | 53   | 40 | 30 | 5.0 | 5.0 | M 10 | 25  | 17      |                          |  |  |
| 90   | 3600                                                   | –      | 4500   | 110                               | –   | –   | 160             | 200 | 104  | 245 | 100    | 62   | 45 | 34 | 5.5 | 6.5 | M 12 | 30  | 40      |                          |  |  |

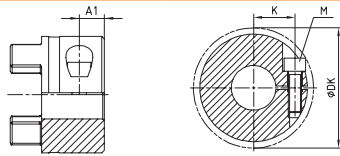
<sup>1)</sup> For selections see page 22 et seqq./other spiders see page 127.

#### Ordering example:

| ROTEX® GS 24  | 98 ShA-GS       | Ø 20                        | 2.5 - Ø24               | 1.0 - Ø20               |
|---------------|-----------------|-----------------------------|-------------------------|-------------------------|
| Coupling size | Spider hardness | Optional:<br>Bore in spider | Hub type<br>Finish bore | Hub type<br>Finish bore |

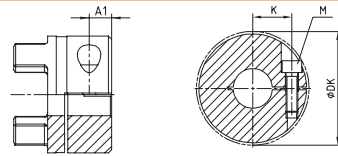
## Types of hubs:

Type 2.0  
Type 2.1



Size 5 to 14  
Type 2.0: single slotted clamping hub **without** feather keyway  
(only for ATEX cat. 3), torque depending on bore Ø  
Type 2.1: single slotted clamping hub **with** feather keyway

Type 2.5  
Type 2.6



from size 19  
Type 2.5: double slotted clamping hub **without** feather keyway  
(only for ATEX cat. 3), torque depending on bore Ø  
Type 2.6: double slotted clamping hub **with** feather keyway

ROTEX® GS

Backlash-free  
servo couplings

## ROTEX® GS standard types - For size 5 to 38 hub material aluminium/for size 42 to 90 hub material steel

| Size | Spider GS <sup>1)</sup><br>torque T <sub>KN</sub> [Nm]<br>for 98 ShA |        |        | D <sub>max.</sub><br>for hub type |     |     |                  | Dimensions [mm] |     |      |     |        |      |    |    |     |     |       |      | Clamping screws DIN EN ISO 4762<br>(ROTEX® GS 5 - DIN EN ISO 1207) |       |                 |  |  |
|------|----------------------------------------------------------------------|--------|--------|-----------------------------------|-----|-----|------------------|-----------------|-----|------|-----|--------|------|----|----|-----|-----|-------|------|--------------------------------------------------------------------|-------|-----------------|--|--|
|      | 98 ShA                                                               | 57 ShD | 64 ShD | 2.0                               | 2.1 | 2.5 | 2.6              | DN              | DH  | DI1  | L   | L1, L2 | M, N | E  | B1 | S   | A   | M     | A1   | K                                                                  | DK    | TA [Nm]         |  |  |
| 5    | 0.9                                                                  | –      | –      | 5                                 | –   | –   | –                | –               | 10  | –    | 15  | 5      | –    | 5  | 4  | 0.5 | 4.0 | M1.2  | 2.5  | 3.5                                                                | 11.4  | – <sup>2)</sup> |  |  |
| 7    | 2                                                                    | –      | 2.4    | 7                                 | 7   | –   | –                | –               | 14  | –    | 22  | 7      | –    | 8  | 6  | 1.0 | 6.0 | M 2   | 3.5  | 5.0                                                                | 16.5  | 0.37            |  |  |
| 9    | 5                                                                    | –      | 6      | 11                                | 11  | –   | –                | –               | 20  | 7.2  | 30  | 10     | –    | 10 | 8  | 1.0 | 1.5 | M 2.5 | 5.0  | 7.5                                                                | 23.4  | 0.76            |  |  |
| 12   | 9                                                                    | –      | 12     | 12                                | 12  | –   | –                | –               | 25  | 8.5  | 34  | 11     | –    | 12 | 10 | 1.0 | 3.5 | M 3   | 5.0  | 9.0                                                                | 27.5  | 1.34            |  |  |
| 14   | 12.5                                                                 | –      | 16     | 16                                | 16  | –   | –                | –               | 30  | 10.5 | 35  | 11     | –    | 13 | 10 | 1.5 | 2.0 | M 3   | 5.0  | 11.5                                                               | 32.2  | 1.34            |  |  |
| 19   | 21                                                                   | 23     | 26     | –                                 | –   | 24  | 24               | –               | 40  | 18   | 66  | 25     | –    | 16 | 12 | 2.0 | 3.0 | M 6   | 11.0 | 14.5                                                               | 46    | 10              |  |  |
| 24   | 60                                                                   | 68     | 75     | –                                 | –   | 28  | 28               | –               | 55  | 27   | 78  | 30     | –    | 18 | 14 | 2.0 | 3.0 | M 6   | 10.5 | 20.0                                                               | 57.5  | 10              |  |  |
| 28   | 160                                                                  | 180    | 200    | –                                 | –   | 38  | 38               | –               | 65  | 30   | 90  | 35     | –    | 20 | 15 | 2.5 | 4.0 | M 8   | 11.5 | 25.0                                                               | 73    | 25              |  |  |
| 38   | 325                                                                  | 365    | 405    | –                                 | –   | 45  | 45               | –               | 80  | 38   | 114 | 45     | –    | 24 | 18 | 3.0 | 4.0 | M 8   | 15.5 | 30.0                                                               | 83.5  | 25              |  |  |
| 42   | 450                                                                  | 495    | 560    | –                                 | –   | 50  | 48               | 85              | 95  | 46   | 126 | 50     | 28   | 26 | 20 | 3.0 | 4.0 | M 10  | 18   | 32.0                                                               | 93.5  | 69              |  |  |
| 48   | 525                                                                  | 590    | 655    | –                                 | –   | 55  | 55               | 95              | 105 | 51   | 140 | 56     | 32   | 28 | 21 | 3.5 | 4.0 | M 12  | 21   | 36.0                                                               | 105   | 120             |  |  |
| 55   | 685                                                                  | 770    | 825    | –                                 | –   | 70  | 70 <sup>3)</sup> | 110             | 120 | 60   | 160 | 65     | 37   | 30 | 22 | 4.0 | 4.5 | M 12  | 26   | 42.5                                                               | 119.5 | 120             |  |  |
| 65   | 940                                                                  | 1055   | 1175   | –                                 | –   | 75  | 75 <sup>3)</sup> | 115             | 135 | 68   | 185 | 75     | 47   | 35 | 26 | 4.5 | 4.5 | M 12  | 33   | 45.0                                                               | 124   | 120             |  |  |
| 75   | 1920                                                                 | –      | 2400   | –                                 | –   | 85  | 85               | 135             | 160 | 80   | 210 | 85     | 53   | 40 | 30 | 5.0 | 5.0 | M 16  | 36   | 51.0                                                               | 147.5 | 295             |  |  |
| 90   | 3600                                                                 | –      | 4500   | –                                 | –   | 90  | 90               | 160             | 200 | 104  | 245 | 100    | 62   | 45 | 34 | 5.5 | 6.5 | M 20  | 40   | 60.0                                                               | 176   | 580             |  |  |

<sup>1)</sup> For selections see page 22 et seqq./other spiders see page 121.

<sup>2)</sup> No T<sub>A</sub> defined (slotted screw)

<sup>3)</sup> From Ø60 keyway opposite the clamping screw

ROTEX® GS HP

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## Review of shaft-hub-connection: Friction torques T<sub>R</sub> [Nm] for hub type 2.0

| Size | Ø3  | Ø4  | Ø5  | Ø6  | Ø7  | Ø8  | Ø9  | Ø10 | Ø11 | Ø12 | Ø13 | Ø14 | Ø15 | Ø16 |
|------|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| 7    | 0.7 | 0.9 | 1.1 | 1.2 | 1.4 |     |     |     |     |     |     |     |     |     |
| 9    |     | 1.6 | 1.9 | 2.2 | 2.6 | 2.9 | 3.2 | 3.5 | 3.8 |     |     |     |     |     |
| 12   |     | 2.4 | 2.9 | 3.4 | 3.9 | 4.4 | 4.9 | 5.4 | 5.8 | 6.3 |     |     |     |     |
| 14   |     |     | 3.1 | 3.6 | 4.2 | 4.7 | 5.2 | 5.7 | 6.2 | 6.7 | 7.1 | 7.6 | 8.0 | 8.5 |

## Review of shaft-hub-connection: Friction torques T<sub>R</sub> [Nm] for hub type 2.5

| Size | Ø8 | Ø10 | Ø11 | Ø14 | Ø15 | Ø16 | Ø18 | Ø19 | Ø20 | Ø22              | Ø24              | Ø25 | Ø28 | Ø30 | Ø32 | Ø35 | Ø38 | Ø40 | Ø42  | Ø45  | Ø48  | Ø50  | Ø55  | Ø60  | Ø65  | Ø70  | Ø75  | Ø80  | Ø85  | Ø90  |      |
|------|----|-----|-----|-----|-----|-----|-----|-----|-----|------------------|------------------|-----|-----|-----|-----|-----|-----|-----|------|------|------|------|------|------|------|------|------|------|------|------|------|
| 19   | 19 | 23  | 25  | 31  | 33  | 35  | 39  | 41  | 42  | 46 <sup>4)</sup> | 49 <sup>4)</sup> |     |     |     |     |     |     |     |      |      |      |      |      |      |      |      |      |      |      |      |      |
| 24   |    | 24  | 26  | 33  | 35  | 37  | 41  | 43  | 45  | 48               | 52               | 54  | 59  |     |     |     |     |     |      |      |      |      |      |      |      |      |      |      |      |      |      |
| 28   |    |     |     | 63  | 67  | 71  | 79  | 82  | 86  | 94               | 101              | 105 | 115 | 122 | 129 | 139 | 148 |     |      |      |      |      |      |      |      |      |      |      |      |      |      |
| 38   |    |     |     |     | 67  | 71  | 79  | 83  | 87  | 95               | 102              | 106 | 117 | 124 | 131 | 142 | 152 | 158 | 165  | 175  |      |      |      |      |      |      |      |      |      |      |      |
| 42   |    |     |     |     |     |     |     | 188 | 197 | 214              | 231              | 240 | 264 | 281 | 297 | 320 | 343 | 358 | 373  | 395  | 417  | 431  |      |      |      |      |      |      |      |      |      |
| 48   |    |     |     |     |     |     |     |     |     |                  |                  | 356 | 394 | 418 | 442 | 478 | 513 | 536 | 558  | 592  | 624  | 646  | 699  |      |      |      |      |      |      |      |      |
| 55   |    |     |     |     |     |     |     |     |     |                  |                  |     |     |     | 456 | 493 | 529 | 553 | 577  | 611  | 646  | 668  | 724  | 778  | 830  | 882  |      |      |      |      |      |
| 65   |    |     |     |     |     |     |     |     |     |                  |                  |     |     |     |     | 499 | 536 | 560 | 584  | 620  | 655  | 677  | 734  | 789  | 842  | 895  | 946  |      |      |      |      |
| 75   |    |     |     |     |     |     |     |     |     |                  |                  |     |     |     |     |     |     |     | 1107 | 1175 | 1242 | 1287 | 1396 | 1503 | 1607 | 1709 | 1810 | 1908 | 2005 |      |      |
| 90   |    |     |     |     |     |     |     |     |     |                  |                  |     |     |     |     |     |     |     |      | 1764 | 1876 | 1985 | 2057 | 2235 | 2409 | 2579 | 2746 | 2911 | 3072 | 3231 | 3387 |

<sup>4)</sup> Clamping hub single slotted with 2-off clamping screws M4 and dimension K = 15, T<sub>A</sub> = 2.9 Nm

RADEX®-NC